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Original Article

Spatial and temporal covariability in early ocean survival of Chinook salmon (*Oncorhynchus tshawytscha*) along the west coast of North America

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Knowledge of the spatial and temporal extent of covariation in survival during the critical ocean entry stage will improve our understanding of how changing ocean conditions influence salmon productivity and management. We used data from the Pacific coastwide coded-wire tagging program to investigate local and regional patterns of ocean survival of Chinook salmon (*Oncorhynchus tshawytscha*) from the Central Valley of California to southeastern Alaska from 1980–2006. Ocean survival of fish migrating as subyearlings covaried strongly from Vancouver Island to California. Short-term correlations between adjacent regions indicated this covariability increased, beginning in the early 1990s. Chinook salmon survivals exhibited a larger spatial scale of variability (50% correlation scale: 706 km) than those reported for other northeast Pacific Ocean salmon. This scale is similar to that of environmental variables related to ecosystem productivity, such as summer upwelling (50% correlation scale: 746 km) and sea surface temperature (50% correlation scale: 500–600 km). Chinook salmon ocean survival rates from southeastern Alaska and south of Vancouver Island were not inversely correlated, in contrast to earlier observations based on catch data, but note that our data differ in temporal and spatial coverage from those studies. The increased covariability in Chinook salmon ocean survival suggests that the marine phase contributes little to the reduction in risk across populations attributable to the portfolio effect. In addition, survival of fish migrating as yearlings from the Columbia River covaried with Chinook salmon survival from the northernmost regions, consistent with our understanding of their migration patterns.

Keywords: Chinook salmon, ocean survival, *Oncorhynchus*, salmon ocean ecology, spatial and temporal covariability, synchrony.

Introduction

Developing robust strategies for the conservation and management of anadromous salmon in a rapidly changing climate requires knowledge of the temporal and spatial scales of the variation in population productivity during their ocean residence (Bisbal and McConnaha, 1998; Schindler *et al.*, 2008; Tolimieri and Levin, 2004). In the northeast Pacific Ocean, for example, both low-frequency, basin-scale variability and local ocean conditions influence salmon productivity (Mantua *et al.*, 1997; Coronado and Hilborn, 1998; Wells *et al.*, 2006; Sharma and Liermann, 2010; Sharma *et al.*, 2013). Coherent decadal scale fluctuations in the harvest of northeast Pacific salmon species have been attributed to decadal scale variability in oceanic conditions [i.e. the Pacific Decadal Oscillation (PDO)], which influences ecosystem productivity in both the Gulf of Alaska and the California Current (Francis and Hare, 1994; Hare and Francis, 1994; Mantua *et al.*, 1997; Hare *et al.*, 1999). Finer scale spatial and temporal variability in Chinook salmon (*Oncorhynchus tshawytscha*) productivity has been linked to strong El Niño Southern Oscillation (ENSO) events or delayed and weak coastal upwelling in the California Current System (CCS) (Johnson, 1988; Kope and Botsford, 1990; Botsford and Lawrence, 2002; Wells *et al.*, 2006). Poor marine conditions in the Gulf of the Farallones led to low early ocean survival of juvenile Chinook salmon emigrating from California's Central Valley in 2005 and 2006 and contributed to the collapse and subsequent salmon fishery closure in California and southern Oregon

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when those cohorts entered the fishery (Lindley *et al.*, 2009). Studying salmon productivity across broad spatial and temporal scales may lend insight into the extent of covariability in productivity and the scale of biophysical processes potentially driving salmon productivity.

A better understanding of spatial and temporal patterns of salmon ocean survival across their geographic range will help managers anticipate the strength of returning runs and cope with uncertainty when developing harvest guidelines and conservation measures (Pearcy, 1992; Mantua *et al.*, 1997; Bisbal and McConnaha, 1998). Chinook salmon typically spend over half their life in the ocean, and survival during the period of early ocean residence is a critical period when the relative strength of salmon cohorts is determined (Pearcy, 1992; Beamish and Mahnken, 2001). The ecological and economic significance of understanding spatial and temporal variation in early ocean survival is highlighted in two aspects of salmon population dynamics. First, the portfolio effect—the buffering effect of within-population diversity of temporal variability of abundance—has implications for both population productivity and persistence (Doak *et al.*, 1998; Moore *et al.*, 2010; Schindler *et al.*, 2010; Carlson and Satterthwaite, 2011). Second, salmon populations are sensitive to (i.e. population abundance becomes more variable with) particular frequencies of random environmental forcing. When populations reach low abundance, due to overfishing or prolonged periods of poor ocean survival rates, this sensitivity increases (Worden *et al.*, 2010; Botsford *et al.*, 2011).

The spatial extent and duration of correlation among survival rates can provide insight into the scale of the oceanographic processes influencing survival (Mantua *et al.*, 1997; Mueter *et al.*, 2002), but must account for differences in life history. Chinook salmon mean age of spawning increases with latitude. Thus, if survival of Chinook salmon in the ocean is similar throughout their range, in order for populations at higher latitudes to replenish their numbers, they should have either higher survival during the early period or higher fecundity to compensate for the additional mortality accrued from spawning at older ages. A congener, coho salmon, spawns at slightly older ages at higher latitudes and has been shown to experience increasing survival with latitude (Teo *et al.* 2009).

The role of physical environmental forcing on Chinook salmon productivity is perhaps less well understood than for other salmon species. Following the mid-1970s regime shift in the northeast Pacific Ocean, salmon production (measured as catch) generally increased in the Gulf of Alaska (GoA), while salmon catches decreased in the CCS—the "inverse production hypothesis" (Mantua *et al.*, 1997; Hare *et al.*, 1999; Mantua and Hare, 2002). This relationship was most clear for coho, sockeye and pink salmon in the GoA. The pattern, however, was less apparent for Chinook salmon (Francis and Hare, 1994; Hare and Francis, 1994; Hare *et al.*, 1999; Botsford and Lawrence, 2002). A comparison of Chinook salmon catches with catches of coho salmon, which primarily spawn at a single age, depicted two distinct population responses to ocean conditions: coho salmon landings increased in the GoA and declined in the CCS after the mid-1970s shift in the PDO, while Chinook salmon catches showed no trend in either system from 1950–1990 (Hare *et al.*, 1999; Botsford and Lawrence, 2002). One possible explanation of the different patterns is that Chinook salmon have a broader spawning age distribution, which could smear the influence of climate on recruitment across multiple cohorts (Kope and Botsford, 1988). A more detailed examination of these catch patterns in the CCS revealed that coho salmon landings from 1950–1990 fluctuated synchronously along the coast, while Chinook landings did not (Botsford and Lawrence, 2002). Recent investigations of Chinook salmon indicate that the influence of local ocean conditions during early ocean residence is important, but a clear consistent explanation remains elusive (Sharma and Liermann, 2010; Sharma *et al.*, 2013). Analysis of wild Chinook salmon stocks showed that their productivity varied with time, but that indicators of ocean state [sea surface temperature (SST), PDO, ENSO] provided little improvement of model fit (Sharma and Liermann, 2010). The influence of the PDO on Chinook salmon productivity appeared to vary with latitude and had a stronger influence on survival in stocks from Washington and Oregon than for those from more northern origins (Sharma and Liermann, 2010). Survival of hatchery fish from California to southeast Alaska was generally negatively related to SST, but covaried at a regional rather than on a basin scale (Sharma *et al.*, 2013). Thus, it is unclear whether the "inverse production hypothesis" applies to Chinook salmon, and the relationship between large-scale climate signals and Chinook salmon productivity remains unclear (Hare *et al.*, 1999; Botsford and Lawrence, 2002).

The coded-wire tag (CWT) dataset provides long-term and broad-scale insights into the ocean ecology of Chinook salmon populations along the west coast of the USA and Canada—a unique dataset for fish harvested in the ocean (Coronado and Hilborn, 1998; Magnusson and Hilborn, 2003; Wells *et al.*, 2006; Trudel *et al.*, 2009; Weitkamp, 2010). Ocean recoveries of CWT-tagged juveniles show that Chinook salmon have consistent year-to-year ocean distributions and that hatchery and wild fish have similar ocean distributions (Trudel *et al.*, 2009; Weitkamp, 2010), thus hatchery-reared and wild Chinook salmon likely experience similar conditions during their ocean residence. In addition, CWTs indicate that spring-run Chinook salmon smolts migrate faster and further north than smolts migrating in their first year of life—fall-run Chinook salmon—and that older fish are recovered farther from their natal streams than younger fish (Trudel *et al.*, 2009; Weitkamp, 2010; Tucker *et al.*, 2011). CWT-based investigations of the spatial and temporal patterns of ocean survival have shown that Chinook salmon survival depends on the ocean conditions (e.g. SST) near the location and timing of ocean entry for migrating smolts and the presence of high quality estuarine habitat (Coronado and Hilborn, 1998; Magnusson, 2002; Magnusson and Hilborn, 2003; Sharma *et al.*, 2013). At larger spatial scales, basin-scale oceanic processes influence the size at recovery of CWT Chinook salmon, with distinct responses for populations originating at different geographic locations (Wells *et al.*, 2006). For example, growth of Chinook salmon from California, Oregon and Washington most strongly correlated with El Niño events during their first ocean winter, while final sizes of Chinook from British Columbia correlated most strongly with El Niño conditions in the year before spawning (Wells *et al.*, 2006).

Here, we focus on the patterns of spatial and temporal covariability in hatchery-reared Chinook salmon survival in the northeast Pacific Ocean, rather than the links between survival and environmental variables across the study area. We examined the CWT data from California's Central Valley to southeast Alaska from 1980–2006. Characteristics at this scale provide insights into the extent that populations share common responses, presumably related to environmental conditions experienced, and hence into

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the scales of physical forcing mechanisms that most strongly influence salmon productivity during the period of highly variable survival after smolts enter the ocean (Pearcy, 1992; Beamish and Mahnken, 2001; Hobday and Boehlert, 2001; Mueter *et al.*, 2002; Beamish *et al.*, 2004). The study region spans three domains of ocean productivity along the western coast of North America based on the physical mechanisms driving ecosystem productivity: (i) the CCS with a coastal upwelling domain south of Vancouver Island, (ii) a transitional domain from 45–50°N where the North Pacific Current bifurcates into the CCS to the south and the Alaska Current to the north, and (iii) a down-welling domain via the northward Alaska Current in the GoA (Mann and Lazier, 1996; Stabeno *et al.*, 2004; Ware and Thomson, 2005). We examined CWT data for Chinook salmon across a range of temporal scales (interannual to decadal) as well as spatial scales (hatchery to regional), which enabled us to investigate how aggregating fine (i.e. hatchery) scale data improved our ability to detect patterns possibly obscured at finer scales while retaining our ability to detect changes in short-term covariability (Levin, 1992; Bakun, 1996). The period over which useable CWT data are available does not include the mid-1970s climatic regime shift observed in the northeast Pacific, but does encompass regime shifts of the late 1980s and the late 1990s, as well as, the 2005 and 2006 cohort failures of fall run Chinook salmon from California's Central Valley that led to costly fishery closures (Peterson and Schwing, 2003; Lindley *et al.*, 2009; Yeh *et al.*, 2011).

Methods

Data

We obtained Chinook salmon CWT release and recovery data from the Regional Mark Processing Center of the Pacific States Marine Fisheries Commission [www.rmpc.org (last accessed 18 January 2012)] and estimated survival of hatchery-reared Chinook salmon released from 1980–2006. We selected 1980 as the initial release year to reduce the influence of variable hatchery, sampling and reporting protocols, although Chinook salmon CWT data collection began in the early 1970s (Hankin *et al.*, 2005; Teo *et al.*, 2009). We chose 2006 as the final release to ensure completeness of recovery data for all cohorts examined. Locations of the hatcheries were obtained from Hobday and Boehlert (2001), Magnusson (2002), Teo *et al.* (2009), Nandor *et al.* (2010) and L. Weitkamp, National Marine Fisheries Service (personal communication).

Data used in the analysis were chosen to minimize irrelevant variability among individual CWT release groups. First, we included only hatcheries with at least five years of releases where each release group had at least 1000 fish. In addition, release groups that were non-experimental, experienced normal hatchery rearing conditions, and had hatchery location, broodstock, and release locations from the same river basin were used (Coronado and Hilborn, 1998; Hobday and Boehlert, 2001; Magnusson, 2002; Teo *et al.*, 2009). To improve geographic and temporal coverage, we included experimental releases for California hatcheries (as did Teo *et al.*, 2009, in a similar study of coho salmon). We characterize CWT release groups as subyearlings (enter ocean same year as hatching) and yearlings (enter ocean one year after hatching) rather than categorizing release groups by "run" or as "stream-type" or "ocean-type". Minimum and maximum size ranges for subyearling and yearling Chinook salmon were set to exclude extremely large and small smolts (subyearling release groups: 25–125 mm and <25 g; and yearling release groups: 50–200 mm and <75 g). In Oregon, releases occurred primarily in the fall compared with the spring and summer for other states; thus, these subyearling release groups were larger than those from other states and we expanded the subyearling weight range to include release groups <75 g. We did not account for differences in release date. A total of 112 hatcheries had sufficient data to include in our analysis (Figure 1).

To facilitate comparison with an earlier study on coho salmon survival, we grouped hatcheries into the same geographical regions used in Teo *et al.* (2009) and added three additional regions: (i) Fraser River; (ii) Puget Sound, Strait of Juan de Fuca, and Strait of Georgia (SOGPS); and (iii) California's Central Valley. This yielded eight regional time-series of subyearling survival and two time-series of yearling survival with at least 20 years of data spanning from southeast Alaska to California's Central Valley. The regions of northern southeast Alaska, Fraser River, northern British Columbia and southern coastal Washington (Figure 1) lacked sufficient data for subyearling or yearling releases for regional level analyses, but were included in hatchery level analyses. Regional analyses facilitated comparison of more complete time-series while maintaining spatial coverage. Hatchery-level analyses captured finer scale spatial patterns (e.g. within river system), but were limited by variable time-series length and consistency. Aggregating data from the hatchery to the regional level may reduce interannual variability of survival rates and, thus, limit the detection of finer scale variability, but this trade-off was necessary to develop time-series of sufficient length.

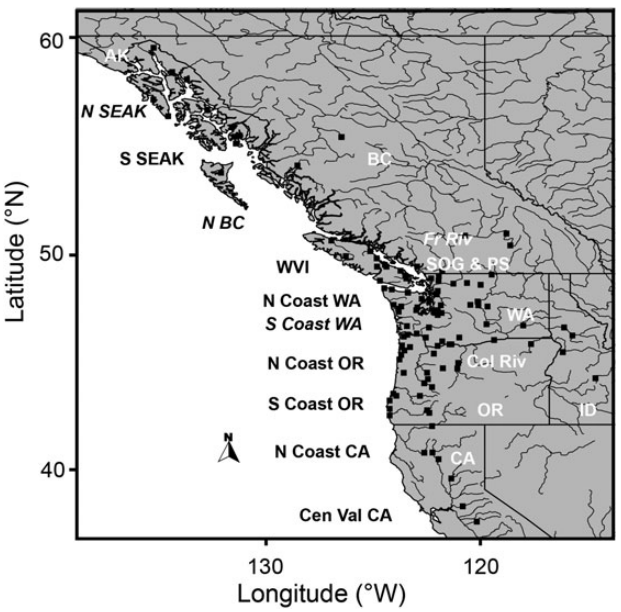


Figure 1. Map of hatchery locations ($n = 112$) and geographic regions used in this analysis. From north to south: (1) northern southeast Alaska (N SEAK); (2) southern southeast Alaska (S SEAK); (3) northern British Columbia (N BC); (4) west coast Vancouver Island (WVI); (5) Fraser River (Fr Riv); (6) Strait of Georgia and Puget Sound (SOG & PS); (7) northern coastal Washington (N Coast WA); (8) south coastal Washington (N Coast WA); (9) Columbia River (Col Riv); (10) northern coastal Oregon (N Coast OR); (11) south coastal Oregon (S Coast OR); (12) northern coastal CA (N Coast CA); and (13) Central Valley of California (Cen Val CA). The regions N SEAK, N BC, Fr Riv and S Coast WA (italics) lacked sufficient data to use in regional analyses.

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We estimated presumed early ocean survival and also accounted for the fact that salmon would be exposed to mortality at older ages for varying amounts of time. Survival was estimated for hatchery smolts released both as subyearlings and yearlings for the period from release to the beginning of fishing the following year. Chinook salmon spawn at multiple ages, typically spanning at least three years of age (Healy, 1991). To account for this variability, we used cohort reconstruction to calculate the number of recoveries for each release group in terms of age-2 (subyearling releases) and age-3 (yearling releases) equivalents (Hilborn and Walters 1992, Chapter 10). Age-a equivalents denote the number of fish estimated to survive their first ocean winter, N_a^*, where a is age, based on reconstructions using assumed ocean survival rates and CWT recovery data for smolts ages a and older (Coronado and Hilborn, 1998; Magnusson and Hilborn, 2003). Survival was calculated as:

$$N_a^* = C_{a+1} + \frac{C_{a+2}}{S_j} + \frac{C_{a+3}}{S_j S_a} + \frac{C_{a+4}}{S_j S_a^2} + \rightleftharpoons, \quad (1)$$

where N_a^* is the number of age-a equivalents from a release group before the opening of the fishery, C_a is the number of expanded CWT recoveries from all fisheries and escapement to the spawning ground for age a for recoveries up to age 8, s_j is the ocean survival from age-a to age-$a+1$ (0.5), and s_a is the ocean survival for all older ages (0.8), similar to previous studies; these values do not influence the observed patterns in survival, but serve only to scale recoveries into a common "currency" for survival estimates (Coronado and Hilborn, 1998; Magnusson and Hilborn, 2003; Hankin and Logan, 2010; Sharma *et al.*, 2013). For subyearling releases, a equals 2 and for yearling release groups a equals 3. The expanded number of returns is based on the specific location, period, and fishery sampled and is calculated by the agency collecting returns. Survival (S_o) was then calculated as:

$$S_o = \frac{N_a^*}{Rel}, \quad (2)$$

where *Rel* is the number of smolts in an individual release group. Average annual survival for each hatchery was calculated as the average of all releases in a given year. Annual survival for each region was calculated as the average of annual survival time-series for all hatcheries within a given region. Survival estimates were log-transformed and normalized ($(x_i - \mu)/s.d.$) to simplify comparison of the time-series with different means and variances.

Analysis

We examined covariability at the regional and hatchery levels using pairwise Pearson's product moment correlation coefficients on log-transformed and normalized survival data. Before conducting correlation analyses, each survival time-series was detrended and the numbers of effective degrees of freedom were calculated to account for intraseries autocorrelation up to lags of ten years (Pyper and Peterman, 1998; Botsford and Paulsen, 2000). Correlation significance was determined assuming that correlation coefficients have a Gaussian distribution with the calculated estimated degrees of freedom and calculating the probability that the observed correlation coefficient is a random draw from the Gaussian distribution. Because the objective of the study was to identify patterns in space and time, rather than test for significant relationships, we did not correct for multiple comparisons. Both detrended and "raw" time-series were used in subsequent analyses.

Analysis of raw time-series allowed examination of the common influence of low frequency signals. At the hatchery level, pair-wise correlations for time-series from all hatcheries with at least ten years of overlapping data were conducted. A similar analysis for regional survival, using regions with at least 20 years of data with at least two release groups, was also undertaken.

To examine survival covariability further at the regional scale, we performed a hierarchical cluster analysis of regional survival time-series. Correlations coefficients from detrended time-series were used to build a dissimilarity matrix (1 − correlation coefficient) and the distance between two nodes was calculated using the average linkage (i.e. the distance between cluster centroids) (McGarigal *et al.*, 2000).

To investigate the spatial correlation scale of hatchery Chinook salmon survival, we estimated the rate of decay for hatchery-level pairwise correlations with respect to the distance between their corresponding ocean entry locations (Myers *et al.*, 1997; Pyper *et al.*, 2001; Mueter *et al.*, 2002; Teo *et al.*, 2009) by fitting a non-linear model of exponential decay:

$$\rho_d = \rho_0 e^{-d/\nu}, \quad (3)$$

where ρ_d is the correlation between the survival time-series of two hatcheries separated by d km based on the point of ocean entry, and ν is the e-folding scale, which measures the distance at which the correlation coefficients are expected to decrease to e^{-1} (36.8%) of the expected mean correlation at a distance of 0 km, ρ_0. To facilitate comparison with previous studies, we also computed the 50% correlation scale, the distance where the correlation coefficient declines to 50% of the value estimated at 0 km. The model was fit using nonlinear least squares weighted by the number of overlapping years of data between the pairs of hatcheries (e.g. Teo *et al.* 2009). The model was also fit with ρ_0 constrained to be 1.0 to facilitate comparison with previous studies; however, trends in the residuals suggested that fitting the ρ_0 parameter was justified in this study. Subyearling and yearling data were evaluated separately. Data from the SOGPS and Fraser River regions were excluded from this analysis because many smolts from these regions do not enter the Pacific Ocean directly.

To examine how patterns of covariability of regional time-series changed with time, we used sliding-window pairwise correlations using detrended regional survival data. This approach provides insight into short-term changes in covariability that may not be apparent in correlation analyses of complete time-series. These could result from an ENSO event or decadal scale variability in local upwelling or SST. Two types of sliding correlation analyses were employed to investigate the extent of spatial and temporal coherence of covariability: (i) comparing the short-term covariability of neighbouring time-series, and (ii) comparing each time-series with selected regional reference time-series. The first approach reveals the coherence of short-term survival fluctuations along a latitudinal gradient, like the subdiagonal of a pairwise correlation matrix. The second approach emphasizes how well the short-term survival pattern of a single region characterizes survival signals across the study range (i.e. how a single element of a correlation matrix compares with all other elements). A sliding window length of five years with no missing data was used, although window lengths of up to ten years were examined. Longer window lengths provided similar results, but yielded fewer comparisons.

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Results

Latitudinal variability in survival

Hatchery-level survival data spanned from ~38–60°N with time-series ranging from 5–27 years. We focused on latitudinal patterns in survival rates at the regional spatial scale (Figure 2a), because patterns in mean survival were more evident at this level than at the hatchery level (Figure 2b). Hatchery-level data from northern British Columbia, Fraser River and the southern coast of Washington were too sparse for regional analyses. No clear trend in mean survival with latitude was apparent at the regional scale (Figure 2a), which contrasts with results for coho salmon that showed a general increase in survival with increasing latitude (Teo *et al.*, 2009). Beginning in the north, mean survival of yearling releases from hatcheries in southern southeast Alaska (S SEAK, 9) was the highest of all regions (> 2.5%) (Figure 2a). The northern-most subyearling release groups (> 54°N) generally had lower mean survival rates than did yearling releases from similar latitudes (Figure 2a). Moving south from the west coast of Vancouver Island to the Columbia River, regional-level mean hatchery survival rates for subyearlings were highest for smolts from the west coast of Vancouver Island (8) and were lowest for subyearlings from the Strait of Georgia and Puget Sound (7) and Columbia River (5) release groups (Figure 2a). Mean survival rates of subyearling smolts from north coastal Washington (6) were slightly higher than those from the Strait of Georgia and Puget Sound (7) and Columbia River (5). Survival rates of Columbia River (5) yearlings were ~1% less than for southern southeast Alaskan yearlings (9) and were slightly higher than those of Columbia River subyearling smolts (Figure 2a). Mean subyearling survival rates of hatchery Chinook salmon were greatest for releases from coastal Oregon (regions 3 and 4), which are released at larger sizes and later in the year than in other regions. These survival rates were approximately twice those observed for subyearling smolts from the southernmost regions in California (1 and 2) and similar to Alaskan yearling releases (2.5%). Mean survival rates of subyearling Chinook salmon were lowest for release groups from coastal northern California hatcheries (Figure 2a and b).

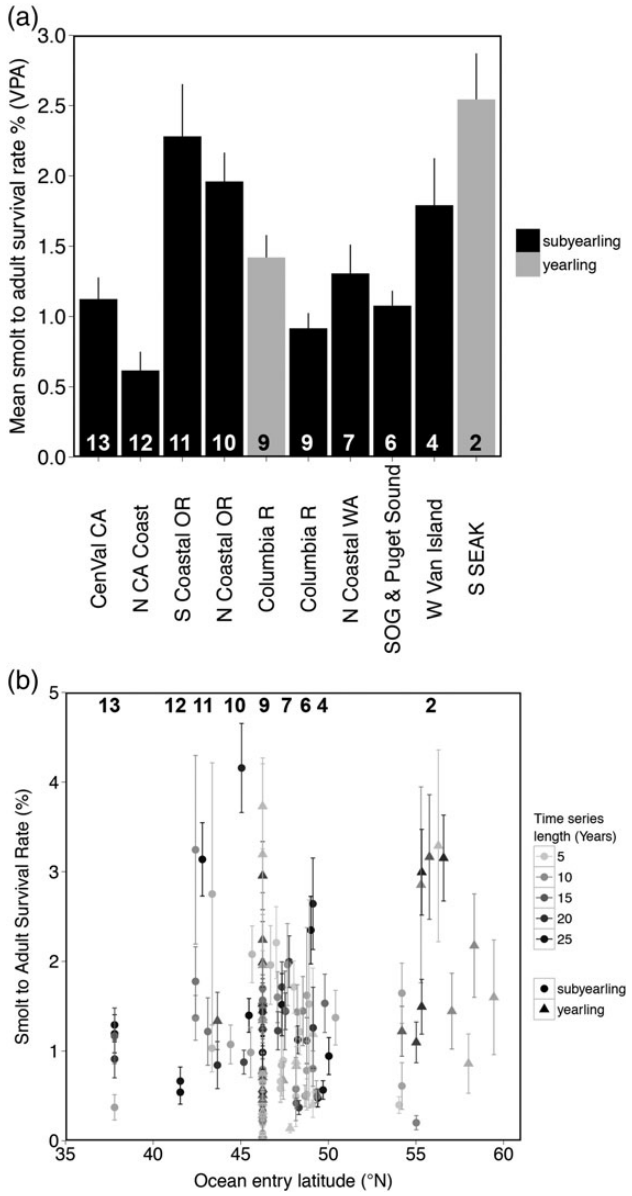


Figure 2. (a) Mean Chinook salmon survival (± 1 se) by geographical region. Black bars indicate subyearling release groups and grey bars indicate yearling release groups. (b) Mean survival (± 1 se) of subyearling (circles) and yearling (triangles) Chinook salmon for each hatchery by latitude of ocean entry. Shading of symbol indicates length of time-series data for each hatchery from 5 (minimum, light grey) to 27 (black) years. To aid comparisons between the regional analyses in this figure and hatchery-level data, regions are identified using the same numbering as in the legend of Figure 1. Note that the region Strait of Georgia (SOG) and Puget Sound overlaps with northern coastal Washington (N Coastal WA) and West Vancouver Island (W Van Island).

Spatial covariability in survival

Covariability across the regional time-series is evident in pairwise correlations (Figure 3a) and hierarchical cluster analysis (Supplementary Figure 3b). In the south, a regional block of high covariability extends from California's Central Valley to northern coastal Oregon with most pairwise correlation coefficients being significant at the 90% level (Figure 3a). This block also appears in the cluster analysis (Figure 3b), as a southern cluster containing subyearlings from California's Central Valley to south coastal Oregon. Moving northward, a second block of covarying regions includes northern coastal Oregon, Columbia River subyearlings, and northern coastal Washington. Although from the same region as the second block of covariability, survival rates of Columbia River yearlings display a different pattern: these fish covaried significantly with release groups from the most northern regions of the study areas (i.e. Strait of Georgia and Puget Sound, west coast Vancouver Island, and southern southeast Alaska), rather than with subyearlings in the neighbouring regions (Figure 3a). The raw and detrended survival time-series had similar patterns of covariability across regions (Figure 3). The results of the regional pairwise correlation analysis are supported by hatchery-level analysis (Figure S1).

Time-series of log-transformed and normalized Chinook salmon survival for smolts released from 1980–2006 highlight three features of spatial and temporal covariability in survival rates among the regional groupings described above (Figure 4). From the west coast of Vancouver Island to the south, the most prominent feature of regional coherence in survival occurred from the late 1990s to the early 2000s, when survival increased until 1999–2001 and then decreased until 2006 in coastal regions

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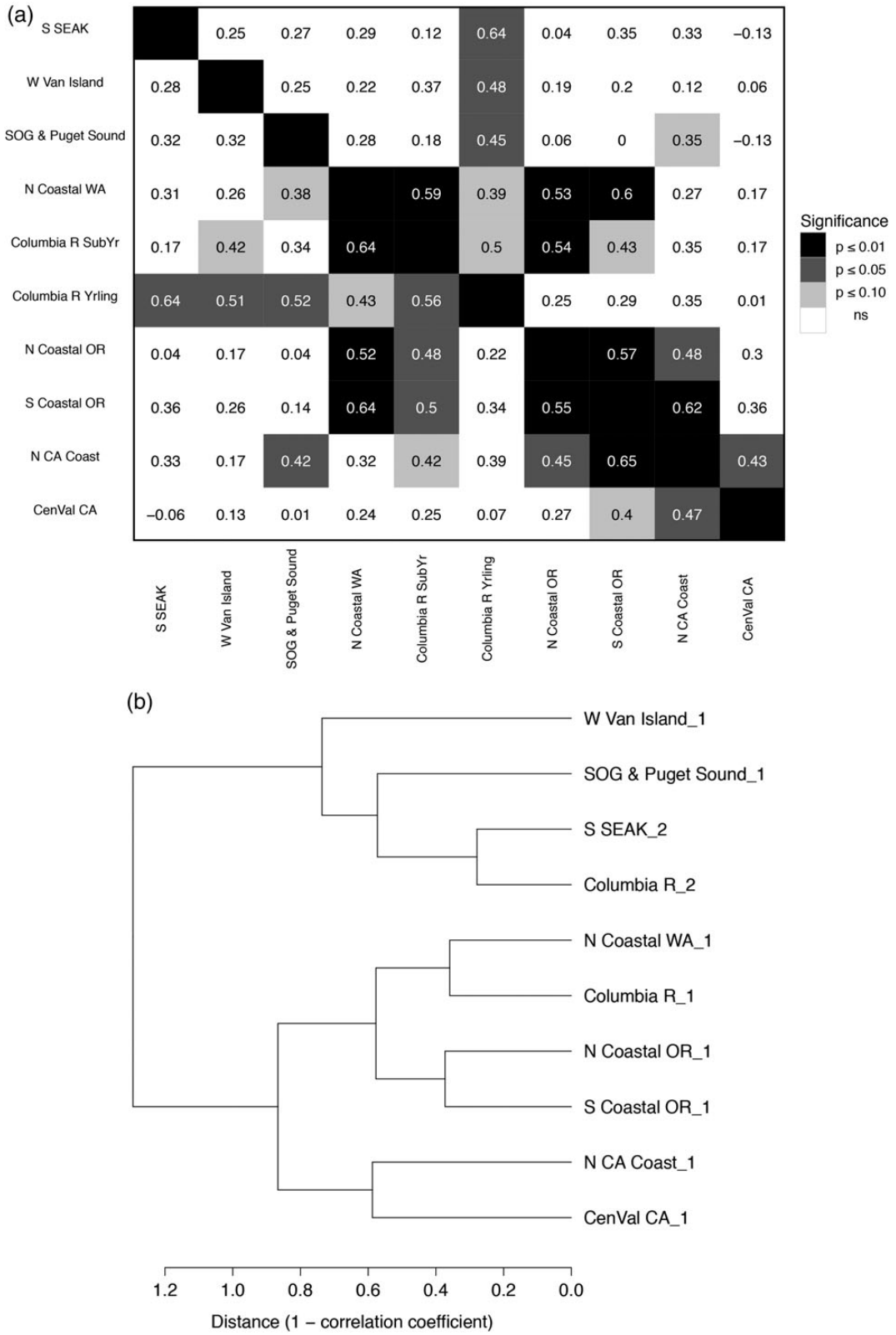


Figure 3. (a) Pairwise correlations of hatchery-reared Chinook salmon survival by region and release stage. All correlations were calculated using log-transformed and normalized survival time-series data. Statistical significance of each correlation is indicated by the shade of the boxes. Results for detrended time-series are above the diagonal, and results for raw time-series are below the diagonal. (b) Cluster dendrogram based on correlations of detrended regional survival time-series. Label suffixes with a 1 or 2 indicate subyearling and yearling releases, respectively.

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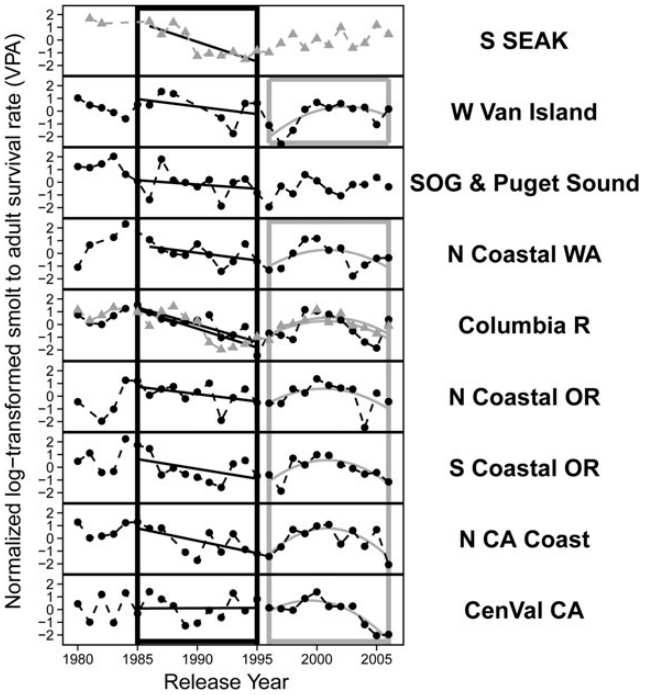


Figure 4. Regional time-series of log-transformed and normalized survival for hatchery Chinook salmon. Black circles (grey triangles) indicate subyearlings (yearlings). The black and grey boxes and solid lines and curves indicate periods of apparent covariability in the time-series. In the black box, the black line represents fit of a linear model from 1985–1995. The grey curve represents the quadratic fit from 1996–2006 in the grey boxes.

(i.e. excluding SOGPS) (Figure 4: grey boxes and dashed curves indicate fit of quadratic model from 1996–2006 for each time-series). Additionally, survival rates of Chinook salmon smolts from California's Central Valley experienced their lowest survival rates in 2005 and 2006—the years that the failed cohorts of Central Valley of fall-run Chinook salmon entered the ocean. A second spatio-temporal feature is a decline in survival from the mid-1980s to the early to mid-1990s north of California's Central Valley (Figure 4: black box and dashed lines indicate fit of quadratic model from 1985–1995 for each time-series). A final feature is that survival rates of yearling Chinook salmon smolts from the Columbia River covaried with those of yearling smolts from southern southeast Alaska until ~2000, when the Columbia River yearling survival began a consistent decline not seen in the southern southeast Alaskan region. Survival in SOGPS peaked before 1990 and then fluctuated without trend.

Spatial correlation scale of survival

The correlation between survival rates of Chinook smolts from different hatcheries decayed slowly with the distance between the ocean entry latitude (Figure 5). We focus on the results for subyearlings, because correlation coefficients of survival for yearling releases did not decay with distance. The maximum distance between ocean entry points was 1915 km for subyearling release groups (filled circles in Figure 5) and 1468 km for yearling release groups (open circles in Figure 5). An exponential decay model with ρ_0 set equal to 1 yielded unacceptable patterns in residuals and poor model fit, and estimated the 50% correlation scale at 217 km (155–291 km, 95% CI). When we estimated both ρ_0 and υ, the model fit was improved (ΔAIC = 71.8). The 50% decorrelation scale was 706 km (440–1437, 95% CI) and the e-folding scale (υ) was 1019 km (635–2074, 95% CI) for the two-parameter model. The correlation at zero distance (ρ_0) was 0.44 in the two-parameter model for the pairwise correlations between hatcheries. These results support the observation that survival of Chinook salmon along the coast of western North America covaries over much of the study range (Figures 2a and 4).

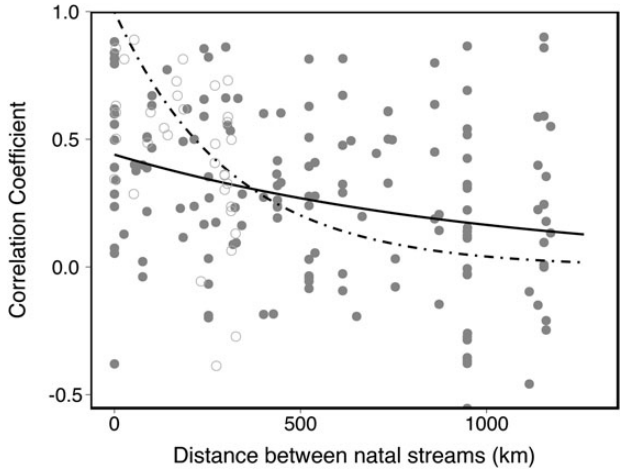


Figure 5. Exponential decay over distance of pairwise correlations between survival time-series from 34 hatcheries of subyearling (grey circles) and 21 yearling Chinook salmon (open circles). Hatcheries from Strait of Georgia and Puget Sound and Fraser River regions were excluded. Lines represent fits to subyearling data only. The solid line represents the fitted model where both ρ_0 and υ were estimated, while the dashed line assumed $\rho_0 = 1$. For subyearlings, the 50% correlation and *e*-folding scales were 706 and 1069 km, respectively.

Temporal variability in spatial covariability of survival

Patterns of covariability in regional survival time-series over shorter temporal scales can highlight periods of change that do not appear in long-term analyses. Short-term correlation analysis of hatchery Chinook salmon survival rates from neighbouring regions showed that the correlation between pairs of neighbouring regions was greatest beginning in the mid-1990s (Figure 6). Limited data in the early study period make it difficult to detect the existence of a long-term increasing trend in covariability between pairs of neighbouring regions. But, in conjunction with the short-term correlation patterns of representative regional time-series with all other regions (south: California's Central Valley; central: Columbia River subyearling and yearling; and north: southern southeast Alaska; Figure 7b and c), strong positive correlations among regions have existed since the early to mid-1990s from California to the west coast of Vancouver Island. The only regional group displaying periods of positive covariabilty with southern southeast Alaska stocks were yearling releases from the Columbia River (Figure 7a). In addition, time-series from other southern regions showed no consistent negative correlations with the southern southeast Alaska region (Figure 7a, b and c).

Discussion

The patterns of spatial and temporal variability in survival of hatchery Chinook salmon observed here provide a basis for considering their contribution to variability in abundance across populations in the study area and developing testable hypotheses regarding the

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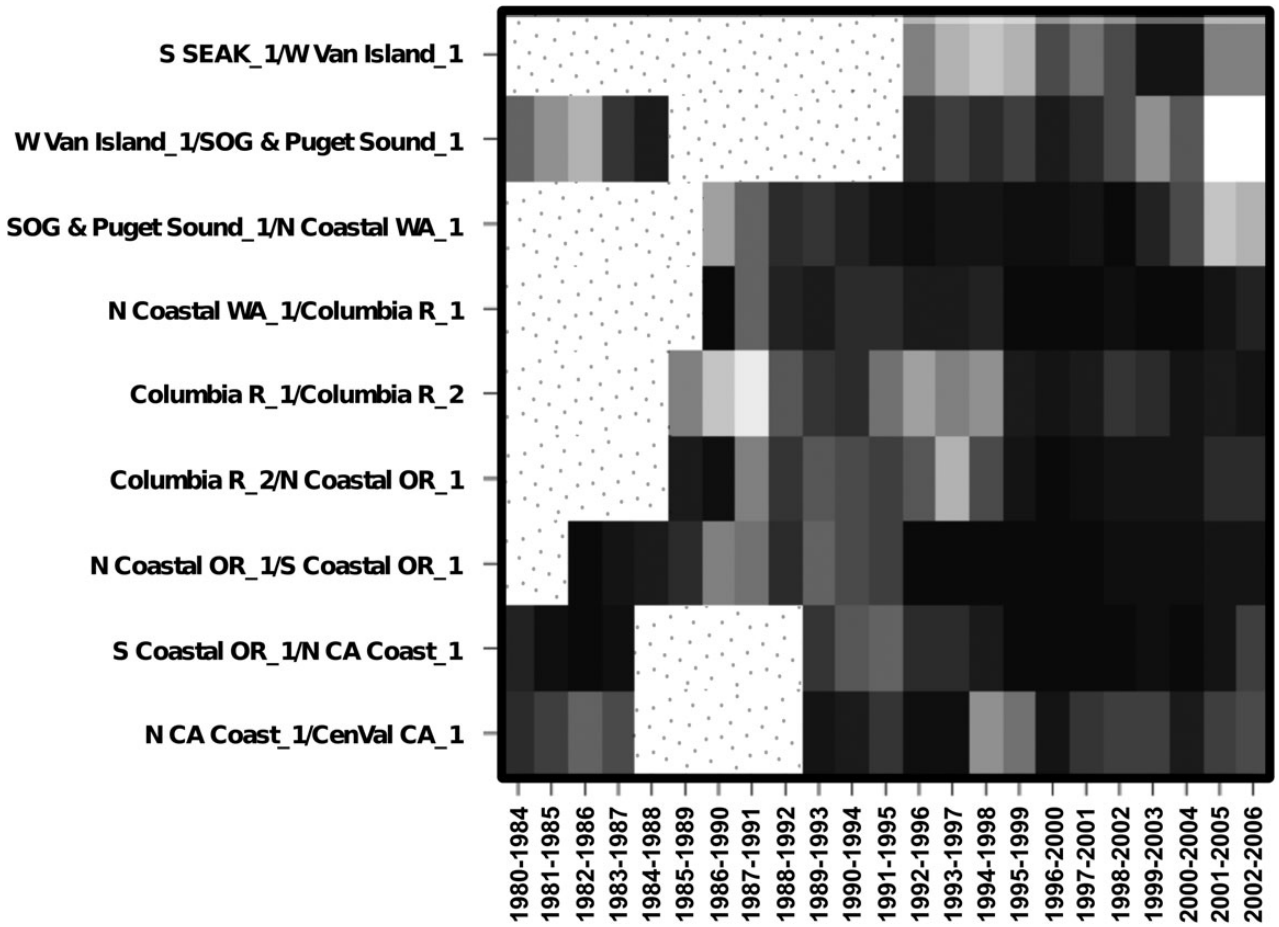


Figure 6. Short-term correlations using a five-year sliding window of Chinook salmon survival for neighbouring (i.e. adjacent) geographical regions over five-year periods from 1980 – 2006. The grey dotted background indicates periods with missing data. Label suffixes with a 1 or 2 indicate sub-yearling and yearling release, respectively.

scale of environmental variability on salmon productivity. In this study, the positive covariability in Chinook salmon survival rates among groups in the CCS, strongest since the early to mid-1990s, differs from patterns of covariability in Chinook salmon catches in the 20th century. Catches of Chinook salmon were less spatially coherent than other salmon species in the northeast Pacific Ocean (Mantua *et al.*, 1997; Hare *et al.*, 1999; Botsford and Lawrence, 2002). The contrasting patterns of spatial coherence between observations of survival and catch data suggest that the lower covariability observed in catch data is not solely a product of survival, but that other factors such as fishing influence spatial patterns of catch. The general positive correlation in survival rates in this study also differs from CWT-based studies of salmon survival that found local conditions (e.g. SST) during ocean entry better explain salmon survival (Coronado and Hilborn, 1998; Magnusson, 2002; Sharma *et al.*, 2013). However, these studies did not report how the strength and patterns of covariance in survival rates have changed with time. Our study specifically examines how hatchery Chinook salmon survival covaries spatially and temporally rather than evaluating how survival covaries with environmental variables (e.g. Sharma *et al.* 2013). In doing so, we detected strong coherence of hatchery Chinook salmon survival beginning in the 1990s. This suggests that survival during the marine phase may contribute little to buffering risk across populations due to the portfolio effect and that recent coastwide patterns of survival might reflect ocean processes operating at the scale of the CCS.

The link between ocean conditions and the observed spatial and temporal patterns of Chinook salmon survival is indirect and depends on biophysical interactions operating across temporal and spatial scales. While we have not evaluated mechanistic relationships between survival and metrics of ocean productivity over multiple spatial and temporal scales, we can surmise possible explanatory mechanisms. The general positive correlation of survival rates observed from the west coast of Vancouver Island and to the south is greatest among neighbouring regions and, as is evident in analyses at shorter time-scales, with the strongest covariability beginning in the early to mid-1990s. The timing of the peak increase in survival across many regions coincides with a state change in the northeast Pacific Ocean in the PDO (late 1990s to early 2000s, Peterson and Schwing 2003). The PDO index shifted to a negative value (i.e. cooler SST) in the late 1990s, but the shift did not lead to a long-term "cool" regime in the 2000s. Rather, since the late 1990s, coastal ocean temperatures have been characterized by 3–5-year oscillations between cool and warm phases (McPhaden, 2012). Though local oceanographic conditions may explain survival of northeastern Pacific salmon (Hobday and Boehlert, 2001; Scheuerell and Williams, 2005; Sharma and Liermann, 2010; Yu *et al.*, 2012; Sharma *et al.*, 2013), local

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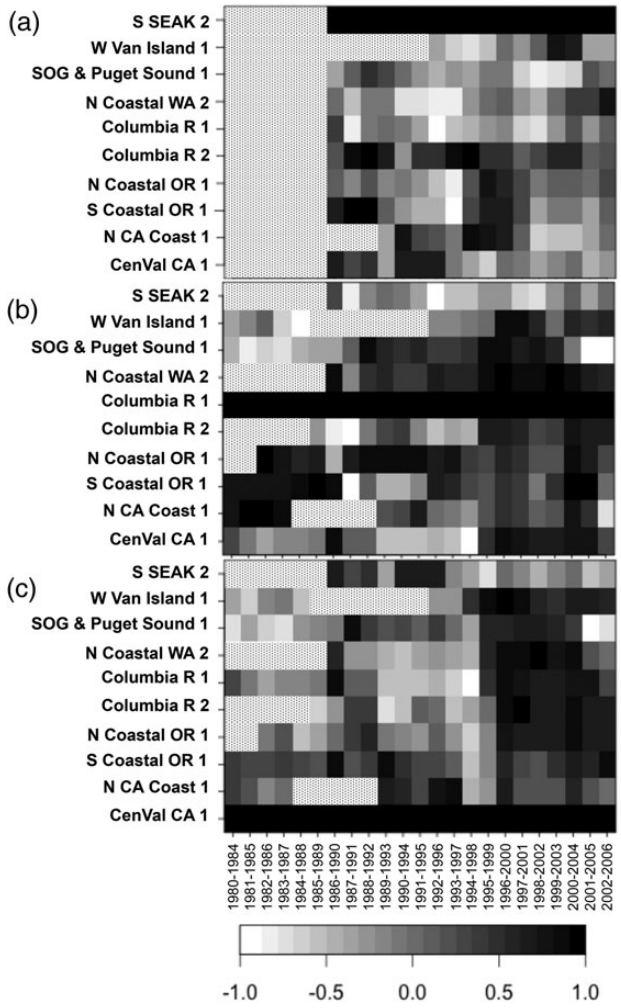


Figure 7. Short-term correlations using a five-year sliding window of Chinook salmon survival in nine geographical regions with respect to a single reference time-series from 1980–2006. Reference time-series used are (a) southern southeast Alaska yearlings (S SEAK_2), (b) Columbia River subyearlings (Columbia R_1), and (**c**) the Central Valley of California subyearlings (CenVal CA_1). The grey dotted background indicates periods with missing data. Label suffixes with a 1 or 2 indicate sub-yearling and yearling release, respectively.

conditions may modify the influence of basin-scale forcing (e.g. El Niño or PDO) on coastal ocean productivity. For example, Sharma *et al.* (2013) note that while local SST best explained survival of hatchery Chinook salmon, SST displayed a "weak" coherence along the coast that can be explained in part by basin-scale ocean–atmospheric processes indexed by the PDO. The spatial correlation scale for hatchery subyearling Chinook salmon (706 km) that is similar to summer upwelling (50% correlation scale: 746 km) and SST (50% correlation scale: 500–600 km) (Mueter *et al.* 2002), suggests that these variables may be important factors. Freshwater conditions, such as stream flow and temperature, experienced by juvenile salmon also contribute variability in survival rates, but the effect of climatic forcing on both the freshwater and marine habitat of salmon may have a similar influence their survival rates (Lawson *et al.*, 2004).

The timing of ocean entry of smolts in relation to the onset of productive conditions (e.g. upwelling) in the CCS is believed to play an important role in determining the survival of salmon cohorts. For example, the years of 2005 and 2006 marked a period of presumed extremely low ocean survival of fall-run (i.e. subyearling) Chinook salmon cohorts entering the ocean from central California, leading to subsequent closures of recreational and commercial Chinook salmon fisheries (Lindley *et al.*, 2009). These years also marked unprecedented complete nest abandonment of Cassin's auklet on Farallon Island, California (Sydeman *et al.*, 2006; Bjorkstedt *et al.*, 2012). Cassin's auklets from Farallon Island have diets and foraging areas similar to those of juvenile Chinook salmon from California's Central Valley. Low prey abundance attributed to delayed upwelling in the coastal ocean in central California were implicated in the low productivity observed in both species in each of these years (Schwing *et al.*, 2006; Sydeman *et al.*, 2006; Barth *et al.*, 2007; Lindley *et al.*, 2009). However, upwelling on the date of ocean entry failed to explain variability in survival in Snake River spring/summer Chinook salmon, where earlier migration of smolts resulted in higher survival rates than those migrating at later dates (Scheuerell *et al.*, 2009). Though upwelling was previously found to predict survival of this population (Scheuerell and Williams, 2005), linking daily upwelling to biological productivity at short time-scales requires accounting for the fact that production at lower trophic levels (phytoplankton, zooplankton) involves delayed responses to upwelling and subsequent relaxation periods (Botsford *et al.*, 2006).

Our CWT-based estimates of survival for hatchery Chinook salmon likely reflect the period of highly variable mortality during the first several months after ocean entry that influences the resulting strength of individual cohorts (Pearcy, 1992; Beamish and Mahnken, 2001; Beamish *et al.*, 2004). Studies of ocean survival indicate that the highest ocean mortality for salmon occurs during the early marine phase and CWT data provide an unparalleled long-term and broad geographic scale perspective into this period for northeast Pacific salmon (Pearcy, 1992). Recently, Welch *et al.* (2011) presented evidence that the critical period of marine survival (for steelhead and sockeye salmon) may occur after the first month of ocean entry (16.5% survival over a 28-day period, compared with 1–4% smolt to adult survival). This is consistent with the hypothesis that the critical period of ocean survival may depend on growth and survival operating over a more protracted period including survival over the first ocean winter (Beamish and Mahnken, 2001) and that our CWT-based survival estimates include this period of high variability in ocean survival. Alternatively, it is also possible that overall survival may reflect conditions experienced by juvenile Chinook salmon during later years of ocean residence, as survival of older ages may be reduced during periods of extreme conditions (Pearcy, 1992). In this study, any variability in survival at older ages was ascribed to survival occurring from smolt emigration through the first year in the ocean.

Previous studies of the scale of 50% spatial correlation scales for salmon and other anadromous species were found to be between those of freshwater and marine fish species and of similar spatial scales to local coastal physical oceanographic variables, with the exception of coho salmon, which covaried over much shorter spatial scales (Myers *et al.*, 1997; Mueter *et al.*, 2002; Teo *et al.*, 2009). Our finding of a 50% correlation scale for hatchery-reared subyearling Chinook salmon, 706 km, is larger than that found for hatchery-reared coho salmon (150 km) and wild pink salmon (399 km) and is more similar to scales reported for wild chum (662 km) and sockeye (532 km) salmon (Mueter *et al.*, 2002; Teo *et al.*, 2009). The spatial correlation scale of Chinook salmon

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ocean survival here differs from the results of Sharma *et al.* (2013), who reported that high (>0.4) correlations occurred most frequently for stock entering the ocean at distances <400 km apart. The differing results can be explained in part by the different methods used in the studies. Sharma *et al.* (2013) tabulated the frequencies of correlation coefficients that exceeded 0.4 across 100-km intervals between natal streams, rather than fitting a statistical model of exponential decay. Other possible explanations include differences in the approaches to selecting release groups and the fact that we included CWT data for California stocks. The larger spatial correlation scale for Chinook salmon contrasts with the results of Botsford and Lawrence (2002), which showed synchronous fluctuations in landings of coho salmon along the California Current system, while Chinook salmon landings covaried over shorter spatial scales. A possible explanation of this effect is that the broader spawning age distribution of Chinook salmon could occlude large-scale environmental effects acting at a specific age (e.g. the age of ocean entry).

This study adds to our understanding of Chinook salmon ocean ecology by identifying broad-scale survival covariabilty in the CCS region, beginning in the early to mid-1990s. Hatchery subyearling Chinook salmon smolts from neighbouring regions along the coasts of western Vancouver Island, Washington, Oregon and California displayed the greatest coherence in survival in our study, while Columbia River yearling smolts correlated most strongly with yearling Chinook salmon from the northernmost regions. The positive correlation in survival rates of Chinook salmon in these regions suggests that marine survival potentially contributes to diminished buffering in variability of abundance across populations through the portfolio effect. This result is of interest because recent analyses reporting increased population synchrony and the consequent increased risk to population persistence predicted by the portfolio effect in both the Snake River and California's Central Valley have attributed it to homogenized freshwater conditions and hatchery fish (Moore *et al.*, 2010; Carlson and Satterthwaite, 2011). A complete examination of the portfolio effect is beyond the scope of this study, but we note that increased correlation in survival due to ocean conditions being increasingly correlated will not be addressed by improving hatchery practices or habitat restoration.

In our study, two distinct groups are apparent along a latitudinal gradient (Figure 3b), one comprised of fish reared in the GoA and the other comprised of fish reared in the CCS. Yearling smolts from the Columbia River, which grouped with the most northern regions, were an exception to the pattern grouping by neighbouring regions was that yearling releases from the Columbia River grouped with the northernmost regions, which is consistent with our current understanding of the more rapid northward migration displayed by yearling Chinook salmon (Fisher *et al.*, 2007; Trudel *et al.*, 2009; Tucker *et al.*, 2011). While we found distinct groupings of CCS and GoA fish, we did not find evidence of an inverse relationship between survival of Chinook salmon from southeast Alaska and the California Current, similar to the results of previous studies of Chinook salmon (Sharma *et al.*, 2013) and coho salmon (Teo *et al.* 2009). In fact, pairwise correlations of regional survival time-series were almost entirely positive (but not necessarily significant). Our study does not span the same period or geographic range as Mantua *et al.* (1997), thus processes relevant to the inverse production hypothesis may still operate over a broader geographic range or on time-scales longer than those in this study.

A clear trend of increasing survival rates with respect to latitude was seen in coho salmon from California to Alaska (Teo *et al.*, 2009), but such a trend is not apparent in Chinook salmon. The lack of a clear trend in Chinook salmon survival rates could be due to both ecological differences and hatchery practices. Because salmon spawning at older ages experience additional years of mortality, our assumption of a constant age-specific survival rate for Chinook salmon after their first ocean year could result in biased estimates of juvenile survival in populations that spawn at older ages (e.g. Alaska). For example, if the assumed age-specific survival rates are higher than the true values, the reconstructed number of *a* year olds (N_a) used would be underestimated, leading to low survival rate estimates, which could potentially mask an increasing trend in survival with latitude as was seen found for coho salmon (Teo *et al.*, 2009). Overall, survival of yearling Chinook salmon tended to be greater than that of subyearling salmon, suggesting that older, larger smolts have higher survival rates. The effect of size can be seen in subyearling Chinook salmon smolts from coastal Oregon, which had the highest ocean survival of subyearling release groups in this study and are released later in the year and at larger sizes than those from other regions. The regions with lowest average survival of subyearling smolts contained river systems with extensive anthropogenic impacts (i.e. Columbia River, Strait of Georgia and Puget Sound, and California's Central Valley), which is consistent with the result that estuarine habitat quality is an important factor for subyearling Chinook survival in their transition from fresh to marine waters (Magnusson and Hilborn, 2003).

The diversity of Chinook salmon life history strategies, particularly within the Columbia River and Sacramento River systems could, in part, explain the low correlation of stocks from common natal streams. For stocks entering the ocean from the same river, correlations theoretically approach one except when (i) there are true differences in survival processes during freshwater residence, and (ii) there is observation error in estimation of the number of returns (Mueter *et al.*, 2002). Difference in freshwater survival may arise from different hatchery locations, release times and release sizes (Bilton *et al.*, 1982; Green and Macdonald, 1987) and from natural diversity in life history variability. Furthermore, observation error in the estimation of the number of returns reduces estimates of the intercept (Mueter *et al.*, 2002). The correlation at zero distance (ρ_0) was 0.44 for the pairwise correlations between hatcheries, but the relative importance of true population differences and observation error is not known. Real differences likely exist in the Chinook smolts originating from common rivers of ocean entry, as 53 of 63 pairwise comparisons from common natal streams were for hatcheries within the Columbia River system. In addition, sampling errors within the CWT recovery program likely contribute to the low ρ_0. A CWT-based analysis of the spatial correlation scale of coho salmon ocean survival found, however, that assuming ρ_0 equalled one adequately fit the data (Teo *et al.*, 2009). A possible explanation of the observed differences between coho and Chinook salmon spatial correlation patterns lies in differences in the freshwater (i.e. longer residence leads to higher ρ_0) and marine (i.e. longer residence leads to lower ρ_0) habitat utilization of these species (Mueter *et al.*, 2002).

The patterns of Chinook salmon survival identified in the analysis of CWT data provided insight into the extent of spatial and temporal covariation during the critical early ocean phase. Examining patterns of covariation across spatial and temporal scales allows us to develop more specific hypotheses about possible underlying physical mechanisms driving the observed covariability patterns. In particular, coherence of Chinook salmon survival rates from neighbouring regions since the mid-1990s

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(Figures 6 and 7) coincides with a state change in the northeast Pacific Ocean (Peterson and Schwing, 2003). This covariation in Chinook salmon survival rates suggests a common population response in the CCS region that is possibly related to changes in the northeast Pacific Ocean that occurred in the late 1990s.

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Supplementary data

The following supplementary data are available at *ICES Journal of Marine Science* online: Supplementary Figure 1.

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